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SYNOPSIS ON

Deep Learning and NLP-Based Real-Time Sign Language Recognition and Contextual Conversion

submitted by

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2025-2026

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ABSTRACT

Sign language serves as a vital communication medium for the Deaf and Hard-of-Hearing (DHH) community, yet regional sign languages such as Kannada Sign Language (KSL) lack robust technological support. This project presents a real-time system for recognizing Kannada Vyanjanagalu (consonant) gestures and converting them into meaningful Kannada text and natural speech. A Convolutional Neural Network (CNN) is trained on the Kannada Vyanjanagalu Sign Language Dataset to classify hand gestures with high accuracy. A lightweight, rule-based NLP module constructs contextual Kannada words, including vattakshara formation and vowel attachment. The system integrates both browser-based speech synthesis and a fallback Kannada Text-to-Speech (gTTS) engine to produce natural audio output. Achieving a training accuracy of **97.6%**, the proposed model demonstrates the feasibility of deep learning–based regional sign translation and highlights its potential to enhance accessibility and inclusion within the DHH community.

CHAPTER-1

INTRODUCTION

Sign language recognition plays a crucial role in bridging communication gaps for individuals with hearing impairments. More than 466 million people worldwide rely on sign language, yet only a small fraction of the population understands it, leading to significant barriers in daily interactions. Regional sign languages such as Kannada Sign Language (KSL) remain underrepresented in modern AI research due to limited datasets and the absence of standardized recognition frameworks.

Advancements in computer vision and deep learning—especially Convolutional Neural Networks (CNNs) and pose-tracking frameworks like MediaPipe—have significantly improved the reliability of sign gesture detection in real time. This project proposes an integrated platform capable of recognizing Kannada Vyanjanagalu gestures from webcam, image, or video input and converting them into readable text and natural-sounding Kannada speech.

The proposed project focuses on creating an end-to-end ASL alphabet recognition system capable of converting gesture inputs into text and natural speech. By leveraging CNN-based spatial feature extraction, NLP-based contextual text formation, and browser-based speech synthesis, the system provides an intuitive, real-time communication platform. Real-world applications include education, healthcare, customer service, and public administration, where automated sign recognition systems can improve accessibility and reduce dependency on human interpreters.

Problem statement

Despite advancements in sign recognition for languages such as ASL or ISL, technical solutions for Kannada Sign Language remain extremely limited. Major challenges include the absence of large annotated datasets, wide gesture variability, inconsistent lighting and background conditions, and high latency in real-time recognition. Moreover, Kannada script requires contextual composition rules for generating meaningful text, making direct mapping from gestures insufficient. There is a need for an integrated, low-latency, AI-driven system that can accurately recognize Kannada consonant gestures, form meaningful text, and produce natural speech to enable smooth communication.

Objectives

- To design and train a CNN-based model for accurate classification of Kannada Vyanjanagalu hand gestures in real time.
- To develop a rule-based NLP module for generating meaningful Kannada text, including vattakshara and vowel composition.
- To integrate a hybrid speech synthesis system using Web Speech API and gTTS for natural Kannada audio output.
- To build a web-based real-time sign-to-text and sign-to-speech platform using webcam input.
- To optimize the model and backend to ensure low latency (150–180 ms) and high accuracy suitable for practical real-world usage.

CHAPTER-2

LITERATURE SURVEY

Paper (Author)	Focus Area	AI Tools Used	Outcomes	Limitations
Boobal et al. [1] (2024)	Real-time sign language and audio conversion for two-way communication	MobileNet, CNN, NLP, Google Speech API	Achieved 94.2% accuracy with real-time responsiveness, multilingual adaptability, and two-way ISL communication	Gesture ambiguity, speech variability, and dataset limitations affect contextual understanding
Bayrak et al. [2] (2024)	ASL recognition using complex-valued deep learning	CZMs, CVDNNs	Achieved 89.01% accuracy on MNIST and 98.67% on Massey dataset; improved precision and robustness	Focused mainly on static gestures; lacks adaptability for dynamic signs
Mukesh Kumar et al. [3] (2024)	Gesture-to-speech translation system for ASL	LSTM, MediaPipe Holistic, Pyttsx3	Achieved 91.63% accuracy; enabled real-time gesture recognition and speech output	Difficulty recognizing facial expressions and multi-modal signs; needs larger datasets
Venkadesh et al. [4] (2024)	Bidirectional ISL communication system (SignChatAI)	CNN, LSTM, Generative AI	Achieved 90–100% accuracy; promoted inclusivity, scalability, and gesture-based communication	Dependent on predefined datasets; limited gesture vocabulary and lacks continuous learning

CHAPTER-3

System Design and Architecture

The System Design and Architecture of real-time sign language recognition and contextual conversion integrates multiple AI technologies to create an efficient and accurate translation framework. The proposed system integrates multiple modules — gesture recognition, NLP-based text construction, and speech synthesis — within a web-based environment.

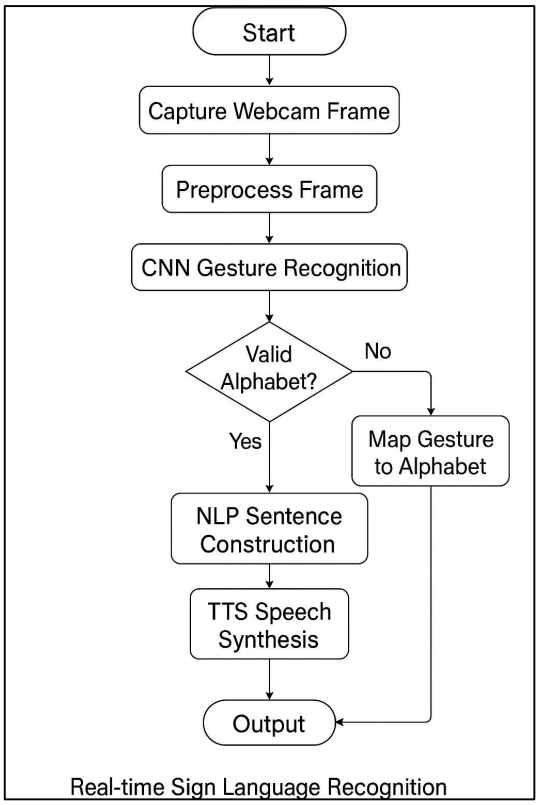

Fig. 1 Workflow Diagram for Real-Time Sign Language Recognition

System Workflow:

1. **Gesture Identification and Recognition**
   - Live video is captured via webcam in the browser, Frames undergo preprocessing: noise reduction, brightness adjustment, background normalization.
   - A CNN model extracts spatial features such as finger contours and palm shapes and the model classifies inputs into 34 Vyanjanagalu.

2. **Sign-to-Text Conversion Using NLP**
   - Predicted labels are mapped to Kannada Unicode characters.
   - Rule-based NLP performs:
     - vowel attachment rules, vattakshara formation,
     - automatic spacing and duplicate suppression and confidence thresholding.
   - Contextual corrections are applied through dictionary-based heuristics.

3. **Real-Time Speech Synthesis**
   - Browser-based Web Speech API converts generated text into speech.
   - Fallback gTTS engine ensures consistent pronunciation and high-quality speech.
   - Speech timings, pauses, and vowel-length adjustments improve clarity.

4. **Web-Based Integration**
   - Frontend built using React.js, HTML, CSS.
   - Backend implemented using FastAPI for live inference.
   - WebSocket streams video frames for real-time processing.
   - Offers visual feedback and immediate gTTS output.

5. **Model Optimization and Deployment**
   - Dataset split: 80% training, 10% validation, 10% testing.
   - Data augmentation improves generalization under different conditions.
   - Training performed on GPU (Google Colab).
   - Achieved 97.6% accuracy with average 180 ms latency.

Technologies Used:

1. Deep Learning & Machine Learning
   - CNN (Convolutional Neural Network) – Gesture classification
   - TensorFlow / Keras – Model training & optimization
   - scikit-learn – Evaluation metrics
2. Computer Vision - OpenCV – Frame capture, preprocessing, normalization
3. Natural Language Processing (NLP)
   - Rule-based Kannada text generation
   - Vattakshara formation, vowel attachment
   - Duplicate suppression & spacing logic
4. Speech Synthesis - gTTS (Google Text-to-Speech) Fallback Kannada voice
5. Web Technologies
   - HTML, CSS, JavaScript – UI/UX
   - FastAPI – Real-time backend inference
   - WebSockets – Low-latency video frame streaming
6. Development Tools
   - Python 3.10, NumPy, Pandas, Matplotlib
   - Google Colab GPU for model training

Outcomes:

1. High Recognition Accuracy - 97.6% accuracy, Precision: 0.98, Recall: 0.97
2. Real-Time Performance
   - ~180 ms latency, 5–6 FPS real-time speed
   - Smooth gesture-to-text and text-to-speech conversion
3. Context-Aware Kannada Text Generation
   - Accurate vattakshara formation
   - Correct Kannada Unicode mapping
   - Automatic spacing and repetition handling
4. Natural Speech Output - Clear Kannada audio through Web Speech API & gTTS
5. Fully Functional Web-Based System - Displays live video, recognized gesture, generated text, and speech output
6. Useful in education, healthcare, government services, and customer support
7. Ready for Future Enhancement
   - Can be extended to swaragalu (vowels), compound letters
   - Can integrate MediaPipe-based tracking & mobile app deployment

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